Andrew Winter

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Curriculum Vitae – October 13, 2026

Overview

I study how the environment surrounding forming stars shapes the protoplanetary discs and planets that emerge from them, connecting star-forming regions, disc evolution, stellar clusters and galactic-scale star formation to observed exoplanet demographics. My work is primarily theoretical, developed in close collaboration with observers and grounded in observational statistics.

Research Positions & Funding Awards

10/26– **Max-Planck-Gesellschaft (MPG) Partner Group Leader** – MPIA partnership, Heidelberg, €100k

05/25– **Royal Society University Research Fellowship**: *'Environmental origins of exoplanet diversity'* – Queen Mary University of London, £1.83M

09/24–04/25 Visiting researcher – Max Planck Institute for Astronomy, Heidelberg

10/23–04/25 **Marie Skłodowska Curie Fellowship**: *'Observational signatures of planet formation in irradiated discs'* – Observatoire de la Côte d'Azur, €195k

10/22–10/23 PDRA – Kinematic signatures of planet formation in protoplanetary discs, Observatoire de la Côte d'Azur

6/22–7/22 Visiting researcher – Chemical inhomogeneities in globular clusters, Institute of Astronomy, University of Cambridge

4/22 **Paris Regional Fellowship** (declined), €130k

3/20–3/22 **Humboldt Fellowship** – *'Planet evolution in dense stellar clusters'*, Heidelberg University, €80k

5/19–3/20 PDRA – *'Photoevaporative disc winds around binary stars'*, University of Leicester

9/18–11/18 DFG funded visiting Ph.D. student – *'Linking galactic-scale star formation physics to protoplanetary disc survival'*, Heidelberg University

Education

2025-ongoing Postgraduate Certificate in Academic Practice (PGCAP), Queen Mary University of London

2015–2019 Ph.D., Institute of Astronomy, University of Cambridge
Graduation date: Oct 2019, Supervisor: Cathie Clarke
Thesis: The influence of stellar birth environment on protoplanetary disc evolution

2011–2015 MMathPhys (Master of Mathematics and Physics), University of Warwick
1st Class Honours, Supervisor: Andrew Levan
Dissertation: Host galaxies of core collapse supernovae

Publications

I have published 80 peer-reviewed articles (18 as first author), with an h-index of 26 and 1920 total citations. A full publication list is available on NASA/ADS.

Conferences & Talks

I have given 20 talks in departments across Europe, the US and China and at 42 international conferences since 2019. This includes 9 invited review talks at large international conferences.

Supervision & Teaching

Supervision	**PhD**: Scarlett Hutton (2026–)*, Oliver Brown (2025–)*, Mika Kontiainen (2024–), Rossella Anania (2022–25), Daniele Fasano (2022–25) **Undergraduate & Masters**: Shreya Harish Kerai (2026–)*, Abubakar Rahimi (2026–)*, Fahham Kurji (2026)*, James Wirth (2023–24)*, Audrey Liu (2026)*, Linling Shuai (2022–23)* *Main supervisor (others: co-supervision or informal)
Teaching	Physical Cosmology Lecturer/Module Co-organiser, QMUL (2025–present); Part II Supervisor – General Relativity & Astrophysical Fluid Dynamics, University of Cambridge (2015–19)

Community & Collaborations

I am an active member of several large collaborations: exoALMA (ALMA large programme), Transition Discs (DFG research unit), EWOCS and XUE. I contribute to the community as a JWST Cycle 4 discussion panelist (previously Cycle 2/3 external panel) and an ALMA distributed peer reviewer, and as a grant reviewer for Fondazione Cariplo, the Humboldt Foundation and STFC, alongside regular refereeing for MNRAS, A&A, AJ, ApJ and Nature Astronomy. I have organised several conferences and workshops, including the PLATO Theory group meeting (QMUL, 2025) and the exoALMA workshop (Nice, 2024).

Observing Time

IRAM	PI (1, Grade B, observations in progress); Co-I (2)
ALMA	Co-I (9), PI (1, Grade C)
JWST	Co-I, 6 awarded programmes, Cycles 1–4 (PIs Bik, Kuhn, Guarcello, Benisty and Ramirez-Tannus)
ESO	Co-I, 9 awarded programmes, Cycles P113–P117

Outreach

2016–26	Public talks and events, including a Think Space lecture (2026) and Cambridge Science Festival talks and a gravity-teaching game, Gravity Hero (2016–18)